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Analysis Sheet for MATSim Scenarios

1. Scenario "Introducing 10 km/h in Dresden's inner city"

[F0] Map of Dresden network links. Yellow links have been assigned a free speed of 10 km/h. Own illustration created with Tramola [S5], based on the Tempo-10 changeset [S3].

This analysis investigates the simulated effects of introducing a large low-speed traffic zone in Dresden's inner city. The scenario is based on the 1% Dresden MATSim model, which simulates traffic in the Dresden area using 1% of the real population and 1% of the real network capacity.

In the Policy Case, 557 selected network links in Dresden's inner city are assigned a free speed of 10 km/h using the Tramola changeset tools [S5]. The policy mainly affects road-based modes using these links, including car, bike, ride and truck traffic. Public transport is not directly affected because it is represented on separate links in the MATSim network.

The analysis focuses on directly affected agents. These are agents who, in the Base Case, complete at least one motorised leg with a route passing through one of the modified links. Their behaviour in the Policy Case is then compared to the Base Case to assess how the Tempo-10 policy changes routing, travel effort and subgroup-specific responses.

Base Case: Dresden 1% scenario without the Tempo-10 changeset.
Link: https://svn.vsp.tu-berlin.de/repos/public-svn/matsim/scenarios/countries/de/dresden/dresden-v1.0/output/1pct/

Policy Case: Dresden 1% scenario with 557 inner-city links reduced to 10 km/h.
Link: https://github.com/RaffaelSo/MATSim-SS26-HA1/tree/main/submission/python-code-and-data/policycase-outputdata

GitHub Repository:
Link: https://github.com/RaffaelSo/MATSim-SS26-HA1

Compiled on 2026-04-20

2. Hypothesis

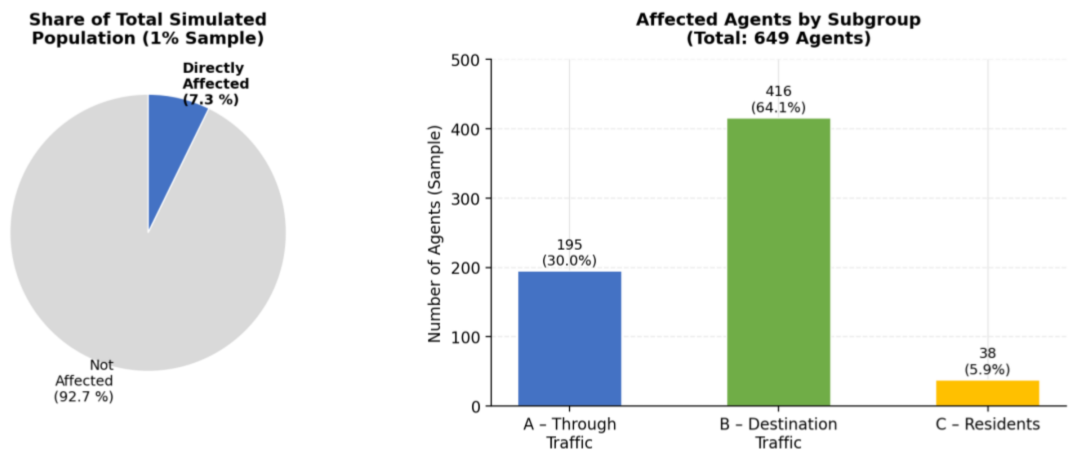
Aspect	Expectation + Reason
Rerouting response	Tempo-10 raises the generalized travel cost of routes through Dresden's inner city. We expect some directly affected agents to avoid the modified links in the Policy Case. This response should be strongest among through traffic, because these agents use the area mainly as a passage and are less tied to activity locations inside the zone.
Travel effort	Reducing free speed to 10 km/h should increase travel time for directly affected agents. Travel distance may change in either direction, because MATSim optimizes generalized travel cost rather than route length: some agents may take detours, while others may choose shorter but slower alternatives.
User-group differences	The policy effect should differ between through traffic, destination traffic and residents. Through traffic is expected to reroute more readily, while destination traffic and residents are expected to remain more often because their activity locations constrain their route choice.

3. Key Observations

1. Observation:

The Tempo-10 policy directly affects a limited but relevant group of agents, most of whom are tied to the zone through activity locations. In the 1% sample, 649 of 8,942 agents (7.3%) use at least one of the 557 modified links in the Base Case [S1], [S3], [S4]. Within this affected group, most agents are destination traffic: 416 agents (64.1%). Through traffic accounts for 195 agents (30.0%), and residents for 38 agents (5.9%) [F1].

Evidence:
Affected agents were identified from Base Case *output_experienced_plans.xml* by checking whether a motorised route passed through one of the modified links. The subgroup classification is based on activity locations on modified links. Summary statistics are reported in [T1] and visualized in [F1]. Data sources: [S1], [S3], analysis script [S4].

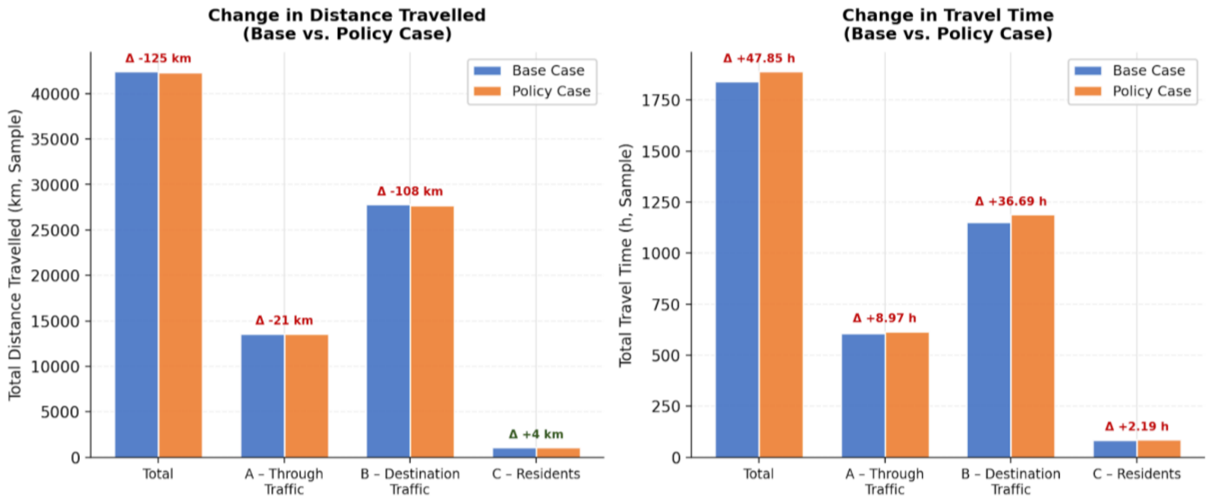


[F1] Directly affected agents overview. Own illustration, based on MATSim Base Case output and the Tempo-10 changeset [S1], [S3]; created with the help of AI (Claude by Anthropic, Model: Sonnet 4.6).

2. Observation:

For directly affected agents, travel time increases while travel distance slightly decreases. Total travel time rises from 1,841.89 h to 1,889.74h in the 1% sample, an increase of 47.85h. At the same time, total distance decreases from 42,430.0 km to 42,305km, a reduction of 124.9km. The number of trips remains unchanged at 3,806, indicating that the policy changes routing and travel effort rather than the number of trips.

Evidence:
Trip-level travel time and distance were compared using Base Case and Policy Case *output_trips.csv*. Summary metrics are reported in [T1] and visualized in [F2]. Data sources: [S1], [S2], analysis script [S4].

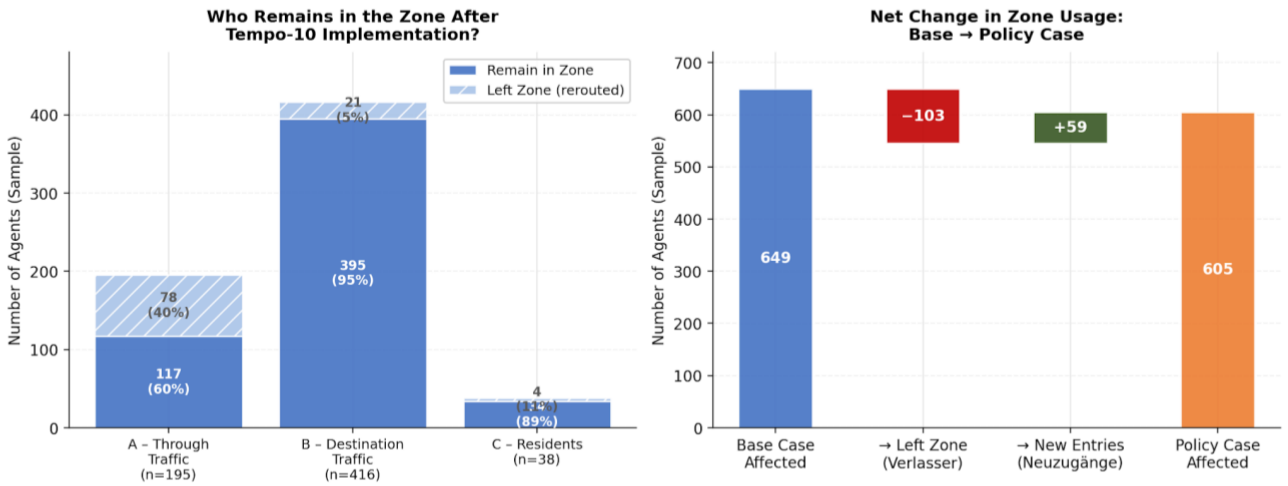


[F2] Distance travelled and travel-time comparison. Own illustration, based on MATSim Base Case and Policy Case trip outputs [S1], [S2]; created with the help of AI (Claude by Anthropic, Model: Sonnet 4.6).

3. Observation:

The Tempo-10 zone produces a clear rerouting response, but the overall deterrent effect is limited as most affected agents still remain in the zone. Of the 649 affected Base Case agents, 546 agents (84.1%) still use the zone in the Policy Case, while 103 agents (15.9%) leave it. The avoidance response is strongest among through traffic: 78 of 195 through-traffic agents (40%) leave the zone. By contrast, destination traffic and residents mostly remain, with 95% of destination traffic and 89.5% of residents still using the zone.

Evidence:
Base Case and Policy Case route-link sequences were compared for the same affected agent IDs. Remainers and leavers by subgroup are reported in [T2] and visualized in [F3]. Data sources: [S1], [S2], [S3], analysis script [S4].



[F3] Zone remainers and leavers. Own illustration, based on MATSim Base Case and Policy Case route-link sequences [S1], [S2]; created with the help of AI (Claude by Anthropic, Model: Sonnet 4.6).

4. Interpretation

The results suggest that the Tempo-10 policy changes route choice, but its deterrent effect is limited by the activity locations of the affected agents. In MATSim, agents choose plans and routes based on generalized travel costs. Reducing the free speed on the inner-city links makes routes through the Tempo-10 zone less attractive, so some agents reroute away from the modified links. This mechanism is most visible for through traffic, where 40% of affected agents leave the zone [T2]. These agents use the area mainly as a passage and are therefore less constrained by fixed activity locations.

At the same time, most affected agents remain in the zone. This is not a contradiction: the affected group is dominated by destination traffic and residents, whose activities or homes are linked to the modified area [T1]. For these agents, avoiding the zone entirely would often require a larger detour or would not be feasible without changing the activity pattern. This explains why 84.1% of all affected agents still use the zone in the Policy Case, even though the modified links have a lower free speed and total travel time for affected agents increases [S3], [T1].

The increase in travel time confirms that the speed reduction creates additional travel disutility for directly affected agents. The slight decrease in total travel distance should not be interpreted as an overall improvement. It indicates that MATSim does not optimize route length alone. Some agents appear to choose shorter but slower alternatives or different routes with lower generalized costs, while the lower free speed still increases total travel time [T1]. Since the number of trips remains unchanged, the policy mainly affects routing and travel effort, not the underlying daily activity plans [T1].

The subgroup differences are therefore central to interpreting the policy effect. Through traffic shows the clearest avoidance response, while destination traffic and residents show strong locational inertia [T2]. The Tempo-10 zone works as a partial deterrent for pass-through movements, but it does not remove most traffic from the area because many affected agents have reasons to remain there.

The decrease in traffic among residents and destination traffic may seem unintuitive at first, because these agents have fixed homes, destinations or activities in Dresden's inner city. However, this can be explained by the definition of the Tempo-10 zone. Only links used by car, bike and truck traffic were reduced to a free speed of 10 km/h, while the public transport network remained unaffected. For agents who are tied to the area but face higher travel costs on the modified road links, switching to public transport is therefore a plausible response [T3].

5. Evaluation

The hypotheses are partly supported. The expected rerouting response is visible, but it is not strong enough to remove most affected agents from the zone. Of the 649 directly affected Base Case agents, 103 agents leave the zone in the Policy Case, while 546 remain [T2]. This supports the expectation that Tempo-10 makes routes through the inner city less attractive. However, the overall deterrent effect is limited because 84.1% of affected agents still use the zone [T2].

The travel-effort hypothesis is supported. Total travel time for directly affected agents increases by 47.85 hours in the 1% sample after the speed reduction [T1]. Travel distance slightly decreases by 124.9 km, which confirms that distance does not necessarily increase when agents reroute [T1]. This result is consistent with MATSim's routing logic: agents respond to generalized travel costs, not simply to physical distance.

The user-group hypothesis is strongly supported. Through traffic shows the clearest avoidance response: 40% of through-traffic agents leave the zone [T2]. Destination traffic and residents mostly remain, with 95% of destination traffic and 89.5% of residents still using the zone [T2]. This confirms that activity locations strongly constrain the reaction to the policy. The Tempo-10 zone is therefore more effective at discouraging pass-through movements than at reducing traffic from agents with destinations or homes in the area.

6. Takeaway

The main takeaway is that a low-speed zone does not affect all users in the same way. In the Dresden inner-city scenario, Tempo-10 creates a measurable rerouting response among through traffic, but most affected agents remain in the zone because they are tied to the area through destinations or homes. This means that the policy works more as a selective deterrent for pass-through traffic than as a general traffic removal measure.

The analysis also shows why MATSim results need to be interpreted through generalized travel costs rather than distance alone. Travel time increases as expected, but total distance slightly decreases, indicating that some agents choose shorter but slower alternatives. For policy analysis, this is important: a speed reduction can change routing and travel effort without fundamentally changing daily activity patterns or the number of trips.

A methodological takeaway is that the definition of "affected agents" strongly shapes the interpretation. We defined affected agents as those who used at least one modified link in the Base Case. This identifies the agents who would have been exposed to the Tempo-10 policy without behavioural adaptation, but it also means that the results describe a targeted subgroup rather than all travellers in Dresden.

The analysis also revealed practical limits of visual inspection tools such as SimWrapper [S6] or VIA. They are useful for exploring spatial patterns and checking plausibility, but they were not sufficient for answering more specific questions, such as which agents leave the zone, which user groups remain, or how travel time changes for directly affected agents. For these questions, additional code-based analysis of MATSim output files was necessary.

Finally, the policy measure is relatively small compared with the full Dresden scenario. Only a limited part of the network is changed, and only a small share of all agents is directly affected. Therefore, large system-wide impacts should not be expected. The value of the analysis lies less in showing a dramatic overall traffic reduction and more in identifying the differentiated behavioural responses of affected user groups.

7. Sources

Repository

GitHub repository: https://github.com/RaffaelSo/MATSim-SS26-HA1
Path convention: All file paths below are relative to the repository root.

Simulation Outputs and Code

[S1] Base Case MATSim outputs: submission/python-code-and-data/policycase-outputdata/base_experienced_plans.xml; submission/python-code-and-data/policycase-outputdata/base_trips.csv

[S2] Policy Case MATSim outputs: submission/python-code-and-data/policycase-outputdata/policy_experienced_plans.xml; submission/python-code-and-data/policycase-outputdata/policy_trips.csv

[S3] Tempo-10 changeset: submission/python-code-and-data/policycase-outputdata/changeset.json, defining the modified network links.

[S4] Analysis script: submission/python-code-and-data/analyse_affected_agents.py, used to identify affected agents and compute subgroup, routing, distance, time and score indicators.

[S5] Tramola documentation: https://docs.simunto.com/tramola/user/getting-started/custom-matsim-in-tramola/

[S6] SimWrapper documentation: https://docs.simwrapper.app/docs/

Tables

[T1] Basic affected-agent metrics: submission/AnalysisSheetHA1/Sources/T1_basic_affected_agent_metrics.csv

[T2] Remainers and leavers by subgroup: submission/AnalysisSheetHA1/Sources/T2_remainers_and_leavers_by_subgroup.csv

[T3] Modal split of affected agents: submission/AnalysisSheetHA1/Sources/T3_modal_split_of_affected_agents.csv

Figures

[F0] Map of Dresden network links with Tempo-10 links: submission/AnalysisSheetHA1/Sources/F0_tempo10_zone_map.jpeg

[F1] Directly affected agents overview: submission/AnalysisSheetHA1/Sources/F1_directly_affected_agents_overview.png

[F2] Distance travelled and travel-time comparison: submission/AnalysisSheetHA1/Sources/F2_distance_and_travel_time_comparison.png

[F3] Zone remainers and leavers: submission/AnalysisSheetHA1/Sources/F3_zone_remainers_and_leavers.png

AI assistance

Claude by Anthropic (Model: Sonnet 4.6) was used to support the analysis code and the creation of the figures. OpenAI Codex (GPT-5) was used to support text editing, document structuring, citation consistency checks and formatting adjustments. All final analytical interpretations and responsibility for the submission remain with the authors.